Fragmented Governance, Density, and Housing Affordability: Evidence from Chicago's Aldermanic Privilege

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Motivation

- Housing prices and rents are skyrocketing in many U.S. cities, including Chicago
  - Limited housing supply in high demand areas is widely accepted as the key driver of these price increases
  - Land use regulations and local control are often blamed for limiting supply
- Most research focuses on formal institutions such as by-right zoning laws or planning commissions
  - But uncertainty and discretion are also key determinates of housing supply and are less studied
- **Chicago**: Unique informal institution of *aldermanic privilege*
  - Offers a unique setting where de facto regulatory environment changes discretely at ward boundaries

What is Aldermanic Privilege?

- **Not written in any law** — Chicago has no city charter
- An informal but extraordinarily powerful norm
- Core feature: **Deference**
  - Other aldermen essentially never override a colleague's decision in their ward
  - "Very few things happen that aldermen don't want to happen." (Zhang, 2011)
  - Encyclopedia of Chicago: "little mayors" ruling wards as "fiefdoms"
- **Chicago gives more power to local politicians than many other U.S. cities**:
  - Houston has no explicit zoning code and centralized development approvals
  - Minneapolis has reduced councilmember discretion in recent years
  - Both cities have seen below-median rent and home price growth compared to peer cities (Hembre et al., 2025; Heckler, 2026)

Aldermanic Privilege is a Hotly Contested Issue in Chicago

CITY HALL NEWS ILLINOIS

Chicago wrongly limited affordable housing with aldermanic prerogative, HUD says

Feds say giving council members "a local veto over proposals to build affordable housing" has meant it's "rarely, if ever, constructed in the majority-white wards that have the least affordable housing."

By Brett Chase | Updated Nov 28, 2023, 5:54pm CDT

May 20, 2019

Mayor Lightfoot Signs First Executive Order to Limit Aldermanic Prerogative

In Her First Official Act as Mayor, Lightfoot Instructs Departments to Begin Process of Ending Aldermanic Prerogative as Part of Broader Ethics Reform

Mayor's Press Office 312.744.3334

Download this Press Release

POLITICS

Developer Moves Forward With Lincoln Park Apartment Complex, Setting Stage for Fight Over Aldermanic Prerogative

Heather Cherone | January 9, 2025, 5:00 am

Controversial Old Town Project With Proposed 44-Story Tower Rejected By Alderman

Ald. Brian Hopkins said he won't support Fern Hill's plan for North and LaSalle. But a new design could be coming.

By Patrick Filbin January 24, 2025 Updated January 27, 2025

Sterling Bay's Marcey Street Development Still Stalled After Heated City Council Exchange

The development would include more than 600 apartments in two towers in Lincoln Park near the Chicago River. Ald. Scott Waguespack opposes the plan in its current form.

By Quinn Myers December 11, 2024

This Paper

Research Question: Does decentralized zoning authority reduce housing supply and raise prices?

Two complementary empirical strategies:

1. *Density at ward borders*: Compare new multifamily construction on opposite sides of ward boundaries
2. *Prices & rents after redistricting*: Event study when census blocks are reassigned to different aldermen

Preview of Results

1. Stricter aldermen lower multifamily density by $13 - 15\%$ near ward borders
2. Census blocks that are redistricted to stricter aldermen see cumulative rent increases of $3 - 5\%$ and home price increases of $\approx 2\%$

Related Literature

Decentralized Urban Governance

- Local control reduces housing supply (Levine, 1999; Mast, 2024)
- Fragmented government $\rightarrow$ underprovision at political boundaries (Bordeu, 2025)

Political Economy of Land Use

- Homevoter hypothesis: owners protect property values (Fischel, 2009; Khan, 2021)
- Incumbent residents oppose density (Hilber and Robert-Nicoud, 2013; Ortalo-Magné and Prat, 2014; Duranton and Puga, 2023)

Border Discontinuity Designs

- School quality (Black, 1999; Bayer et al., 2007)
- Housing regulation and local government (Turner et al., 2014; Kulka et al., 2022; Monarrez and Schönholzer, 2023)

A "Time-to-Build" Model of Regulatory Stringency

Key idea: Regulatory environment $\rightarrow$ effective delay $T(s)$ $\rightarrow$ discounts developer revenue

Developer's problem. A project chooses density $q \geq 0$ with cost $C(q) = F + cq + \frac{\gamma}{2}q^2$.

Revenue arrives upon completion, discounted by regulatory delay:

$$\max_{q \geq 0} \quad e^{-rT(s)}\, p\, q \; - \; C(q)$$

where r = discount rate, $T(s)$ = time-to-build under regulatory environment s

Optimal density (interior solution):

$$q^*(T, p) = \frac{e^{-rT(s)}\, p - c}{\gamma} \qquad \text{if } e^{-rT(s)} p > c, \text{ else } q^* = 0$$

Sufficient statistic: $rT(s)$ summarizes the regulatory wedge — captures delay, redesign, negotiation, financing frictions, and regulatory uncertainty

Testable Predictions

Core comparative static: $\partial q^* / \partial T(s) < 0$, which implies:

1. **Lower density** — completed projects have fewer units → Border-pair FE
2. **Fewer projects** — more sites flipped from "build" to "don't build" → Border-pair FE
3. **Higher prices/rents** — lower aggregate supply $Q = \sum_i q_i^*$, $p'(Q) < 0$ → Event study

Mapping to data:

- $T(s)$ proxied by observed permit-to-completion times
- Aldermanic strictness score $\approx$ alderman-specific component of $T(s)$
- Cross-boundary variation: different aldermen $\Rightarrow$ different $T(s)$ on each side

The Challenge: Measuring Alderman Strictness

- Alderman preferences/frictions are latent: need to infer from observed behavior
- **Idea**: Use permit processing times as a **proxy** for regulatory strictness
  - Data: Universe of Permits in Chicago (2006-2025) from Department of Buildings
  - Longer average processing time $\rightarrow$ more back-and-forth, more friction
- Focus on **high-discretion permits**: new construction, renovation, demolition

Strictness Score Construction

1. **Residualize** processing times on ward characteristics + year × month FEs
   - Controls for baseline differences across wards (income, demographics, etc.)
2. **Estimate alderman fixed effects** on residualized processing times

$$\widehat{\epsilon}_{it} = \mu + \delta_a + \eta_{it}$$

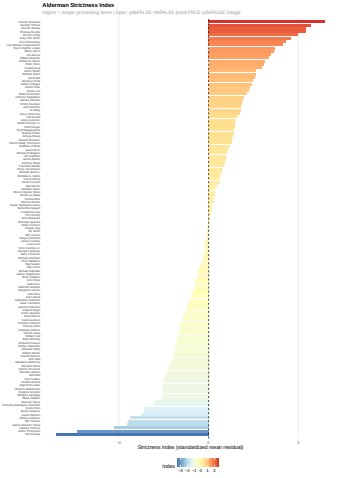
3. **Empirical Bayes shrinkage** to reduce noise (Kline et al., 2022, 2024) then standardize to mean 0, SD 1

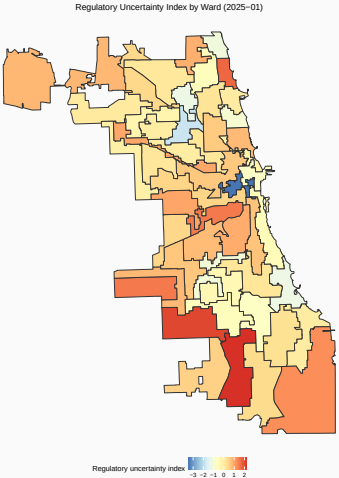
Interpretation: High score = alderman associated with longer processing times

Caveat: This is a *proxy*: likely understates true variation in bureaucratic frictions

- Misses projects that never apply or are discouraged early

Alderman Regulatory Uncertainty Index

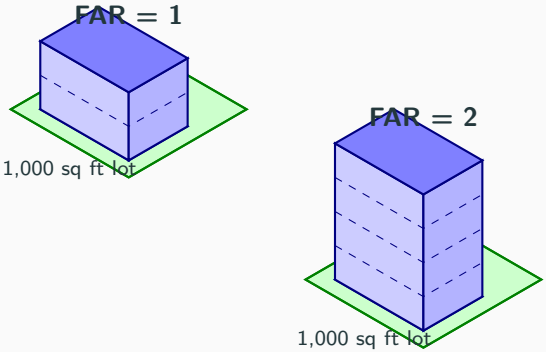




Regression

Score Validation

Density Outcomes: Floor Area Ratio (FAR)



$$\text{FAR} = \frac{\text{Total Building Floor Area}}{\text{Lot Area}}$$

FAR limits development intensity

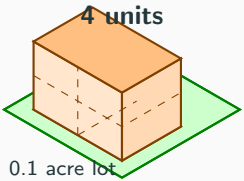
Same lot, different allowed density:

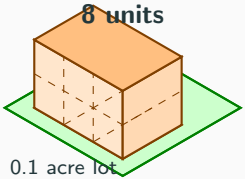
- FAR = 1: Can build 1,000 sq ft
- FAR = 2: Can build 2,000 sq ft

Chicago zoning examples:

Zone	Max FAR	Typical Use
RS-3	0.9	Single-family
RT-4	1.2	Two-flat
RM-5	2.0	Small apartment
RM-6	4.4	Mid-rise

Density Outcomes: Dwelling Units Per Acre (DUPAC)





DUPAC: 40 vs. 80 units/acre (same building volume!)

DUPAC captures population density

$$\text{DUPAC} = \frac{\#\ \text{Dwelling Units}}{\text{Lot Acreage}}$$

Why both measures?

- Same FAR, different DUPAC
- **Developers** care about FAR (total rentable space)
- **Neighbors** care about DUPAC (congestion, parking, schools)

Data: New Multifamily Construction

Source: Cook County Assessor parcel data, 2006-2025

Sample: Multifamily buildings with 2–100 units

- **Single family**: Majority of construction but less variation in density
- **Large developments** (100+ units): Planned Development process with more city oversight

Why multifamily matters:

- Only ∼19% of new construction but a dispraportionate share of total housing supply

Summary statistics:

	All	MF Only
Avg FAR	1.22	2.32
Avg Units	6.65	31.3
Avg Stories	2.27	2.75
Avg DUPAC	28.5	74.5
N	13,236	2,469

Identification Strategy: Border Discontinuity

Idea: Compare new multifamily construction on opposite sides of ward boundaries

- Focus on parcels **within 500 feet** of border
- Use stringent border-pair $\times$ year FEs to remove unobserved heterogeneity

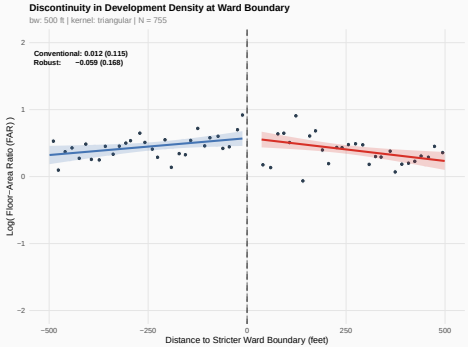
Identifying variation: At the same border in the same year, does the stricter side have less dense construction?

Two specifications:

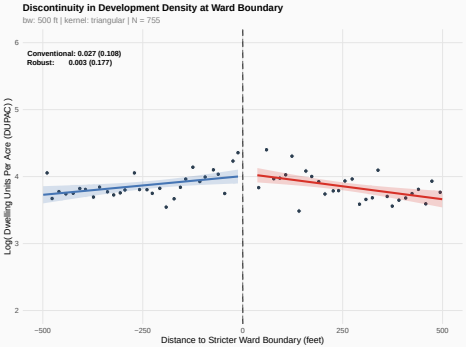
- **Visual RD**: Descriptive, visual evidence of discontinuities at ward borders
- **Regressions**: Regression with continuous strictness measure and stringent FEs

Visual Evidence: Density Discontinuity at Ward Borders

Log(FAR)



Log(DUPAC)



Notes: Stacked borders; lenient (left) vs. strict (right). 500 ft bandwidth.

▸ Placebo Boundaries ▸ Functional Form ▸ Donut

Main Results: Border-Pair Fixed Effects

Specification: $\ln(Y_i) = \beta \cdot \text{Strictness}_i + \alpha_{\text{border} \times \text{year}} + \varepsilon_i$

Sample: Within 500 ft, 2-100 units. Standard errors clustered at border-pair level.

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Strictness Score	-0.15**	-0.10	-0.11*
	(0.06)	(0.06)	(0.06)
Observations	657	657	657
Dep. Var. Mean	1.88	57.81	10.14
Ward Pairs	106	106	106
Ward-Pair × Year FE	✓	✓	✓

Interpretation: A 1 SD ↑ in uncertainty is associated with lower FAR/Units; DUPAC is less precisely estimated

▸ 250ft ▸ 1000ft

From Density to Prices

So far: Stricter aldermen $\rightarrow$ lower multifamily density

Next question: Does restricted supply affect housing costs?

Exploit redistricting as natural experiment

- Ward boundaries redrawn after each decennial census (2015, 2023)
- Some census blocks reassigned to a new alderman while neighboring blocks do not
- Compare prices before vs. after reassignment

Event Study Design

Intuition: When a block switches from lenient $\rightarrow$ strict alderman,
expected future supply $\downarrow \Rightarrow$ prices should $\uparrow$

Specification:

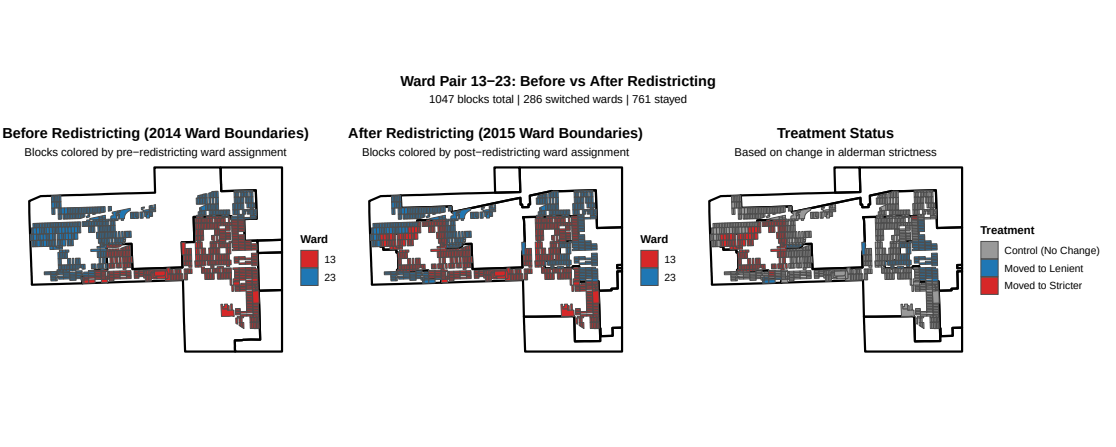
$$\ln(P_{it}) = \sum_{\tau \neq -1} \beta_\tau \cdot \mathbf{1}[t = \tau] \cdot \Delta\text{Strictness}_i + X_{it}'\gamma + \alpha_{ps} + \lambda_{pt} + \varepsilon_{it}$$

- $\Delta\text{Strictness}_i = \text{Strictness}_{\text{dest}} - \text{Strictness}_{\text{origin}}$
- X_{it}: Hedonic controls (sqft, beds, baths, building type/age)
- α_{ps}: Border-pair $\times$ origin-side FE
- λ_{pt}: Border-pair $\times$ year FE

Key assumption: Parallel trends within each border-pair $\times$ side cell

Sample: Within 1,000 ft of boundaries; triangular weights $w_i = 1 - \frac{d_i}{1000}$; SEs clustered by block

Identification: Example from Wards 13 & 23



Blocks reassigned between wards with different strictness scores provide identifying variation.

▸ Citywide Maps

Event Study Data

Rental Listings

- Scraped rental listings from Renthub, 2014-2025
- Asking rents
- **Stacked design**: 2015 + 2023 cohorts (implementation date)
- N $\approx$ 3.9 million near borders

Home Sales

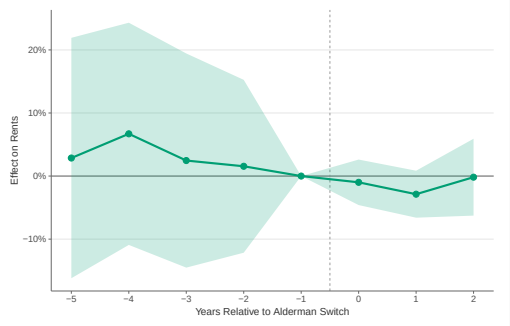
- Cook County Assessor Data from Illinois Department of Revenue, 2007-2025
- Arms-length transactions only
- **Stacked design**: 2012 + 2022 cohorts (announcement date)
- N $\approx$ 78,000 near borders

Sample: Transactions/listings within 1,000 ft of ward boundaries

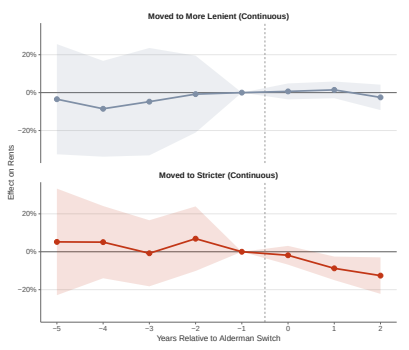
▸ Full Treatment & Control Details

Effect on Log(Rents)

Aggregate Treatment Effect



Stricter vs. Lenient



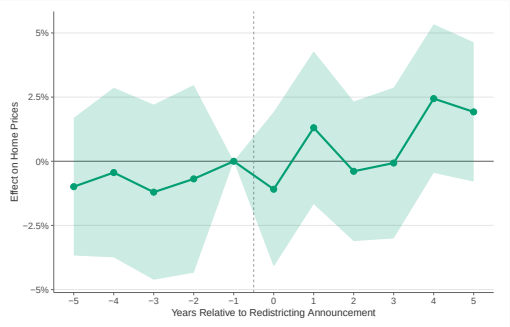
Rent responses are muted and imprecise under the uncertainty index.

Split estimates are directionally informative but noisy.

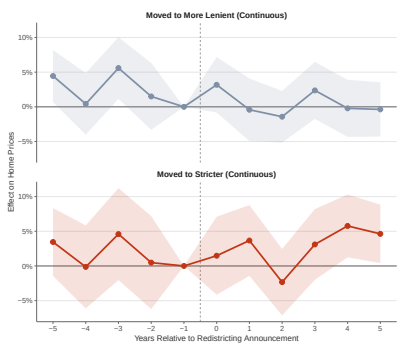
▸ 500ft ▸ No Controls

Effect on Log(Home Prices)

Aggregate Treatment Effect



Stricter vs. Lenient



Home prices show small positive effects with wide confidence intervals.

▸ 500ft ▸ No Controls

Difference-in-Differences Estimates

Log(Rents)

	No Hedonics	Hedonics
Post × ΔStrict	-0.016	-0.007
	(0.032)	(0.021)
N	3.88M	3.88M
Dep. Var.	Mean: $2,540 / Median: $2,345	

⇒ Point estimates for rents are near zero/slightly negative

Log(Home Prices)

	No Hedonics	Hedonics
Post × ΔStrict	0.009	0.010
	(0.008)	(0.007)
N	77,639	77,609
Dep. Var.	Mean: $340K / Median: $265K	

⇒ Point estimates for prices are positive but imprecise

Border-pair × side FE and border-pair × year FE. Triangular kernel, 1000 ft bandwidth.

▸ Covariate Balance

Discussion & Policy Implications

Core problem: Aldermen don't internalize citywide benefits of density

- Agglomeration spillovers extend beyond ward boundaries
- Negative externalities (congestion) are local
- $\Rightarrow$ Systematic under-provision of housing relative to the social optimum

Policy alternatives to centralize decision-making:

1. **Transfer decision-making authority**: Shift decision-making to the citywide level (i.e. mayor, zoning board)
2. **Citywide preemption**: Override local vetoes for certain housing types (e.g., CA SB-35, MA Ch. 40B, federal H.O.M.E. Act proposals)
3. **Less aldermen**: Reduce number of wards to increase average ward size and internalize more spillovers

Conclusion

Summary of findings:

1. Aldermen vary substantially in how they exercise zoning discretion
2. Stricter aldermen $\Rightarrow$ 10-16% lower multifamily density at borders
3. Stricter aldermen $\Rightarrow$ 3-5% higher rents, 2% higher home prices

Implications:

- Decentralized zoning creates real welfare costs
- Within-city fragmentation matters, not just across municipalities
- Reforms should consider who bears costs vs. who makes decisions

Future work:

- Structural estimation of agglomeration spillover decay
- Welfare analysis of alternative governance structures

Thank you!

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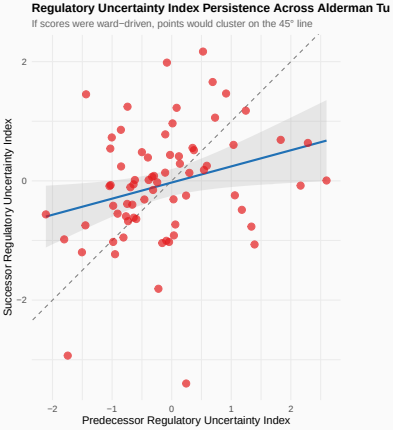
Predecessor-Successor Score Correlation



Ward effects explain 56% of score variance (44% within-ward).

Predecessor-successor correlation: $\rho = 0.27$.

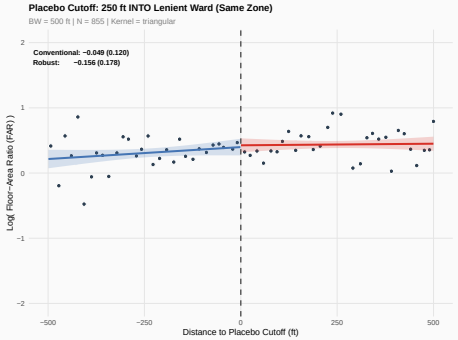
Back

Permit Processing Time Residualization

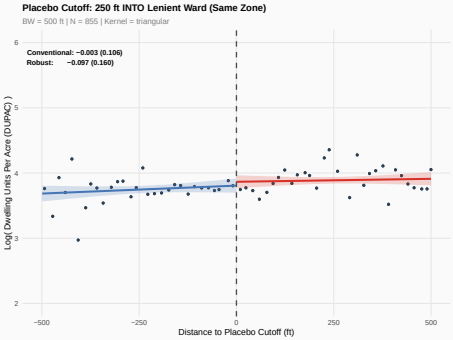
	Log Processing Time (1)
Median HH Income	$-1.37 \times 10^{-6*}$
	(7.5×10^{-7})
Share Black	-0.045
	(0.074)
Share Hispanic	0.041
	(0.051)
Share White	0.397***
	(0.082)
Homeownership Rate	-0.050
	(0.075)
Share Bachelor's+	-0.080
	(0.067)
Population	$-6.21 \times 10^{-6***}$
	(1.28×10^{-6})
Dist. to CBD (km)	-0.007***
	(0.002)
Lakefront Share (legacy)	-1.02***
	(0.102)
CTA Stations (legacy 800)	-0.011***
	(0.002)
Observations	139,787
R^2	0.05735
Month FE	✓

Placebo: Boundary Shifted 250ft into Lenient Side

Log(FAR)



Log(DUPAC)



No discontinuity at placebo boundary.

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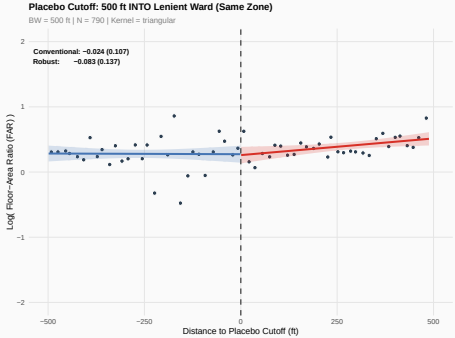
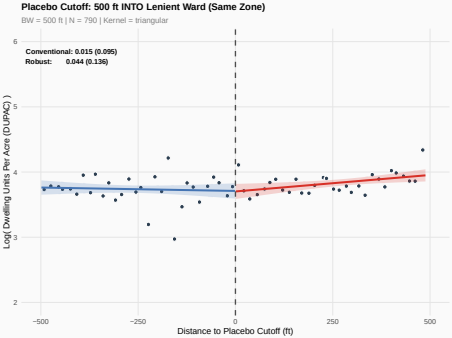
Placebo: Boundary Shifted 500ft into Lenient Side

Log(FAR)



Log(DUPAC)



No discontinuity at placebo boundary.

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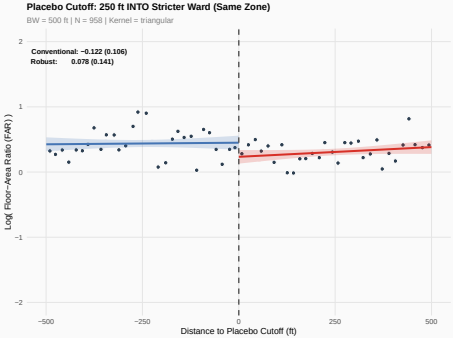
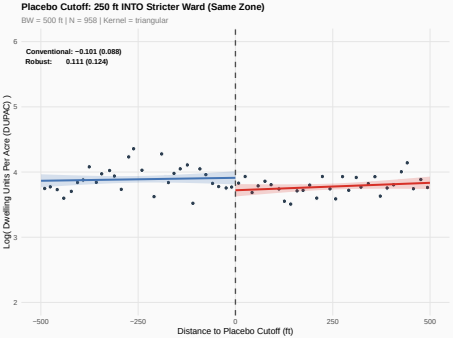
Placebo: Boundary Shifted 250ft into Strict Side

Log(FAR)



Log(DUPAC)



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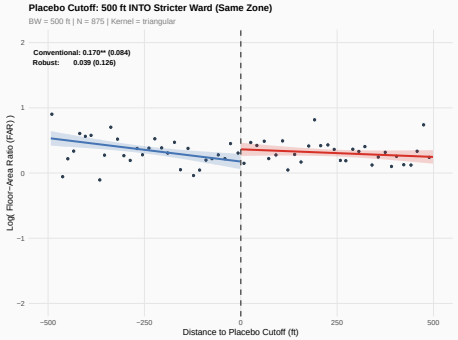
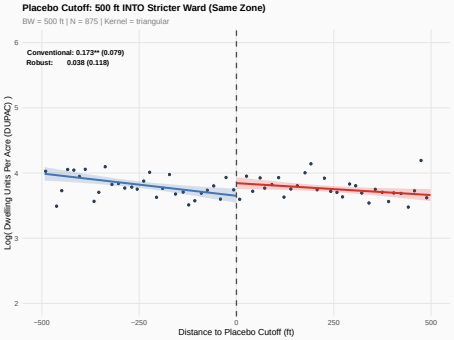
Placebo: Boundary Shifted 500ft into Strict Side

Log(FAR)



Log(DUPAC)



No discontinuity at placebo boundary.

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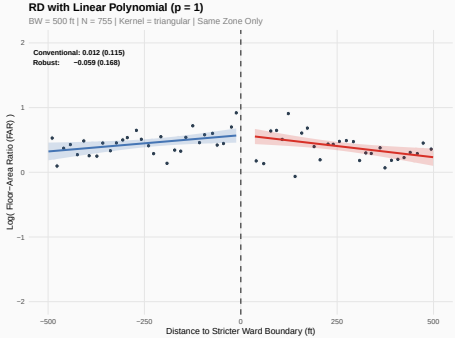
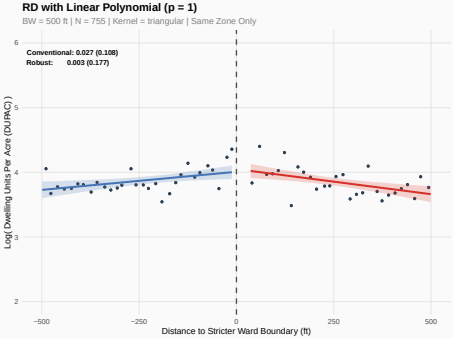
Functional Form: Linear

Log(FAR)



Log(DUPAC)



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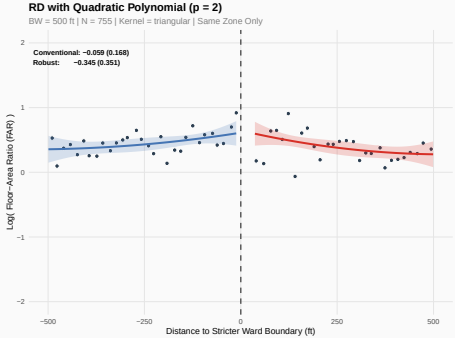
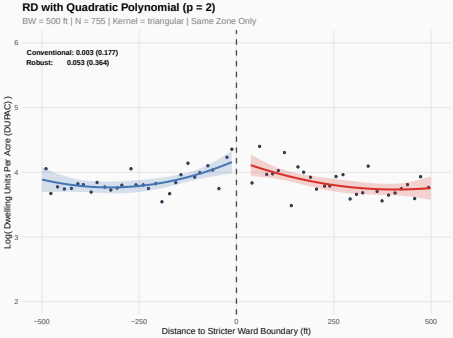
Functional Form: Quadratic

Log(FAR)



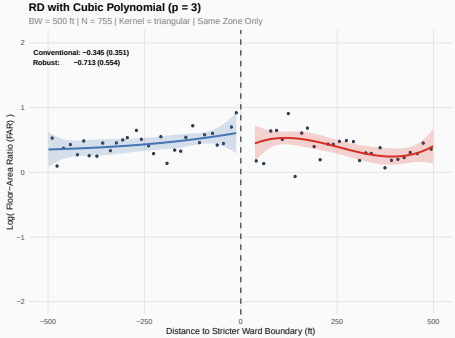
Log(DUPAC)



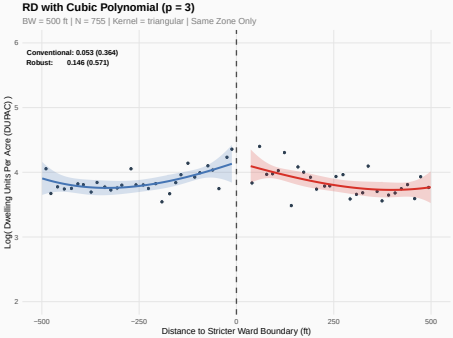
Back

Functional Form: Cubic

Log(FAR)



Log(DUPAC)



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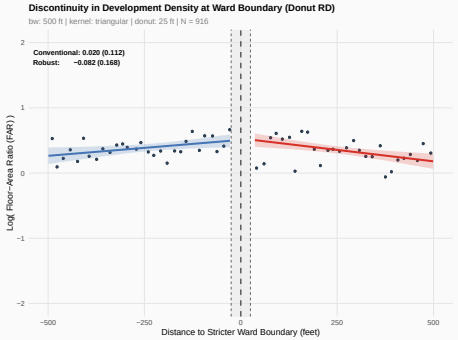
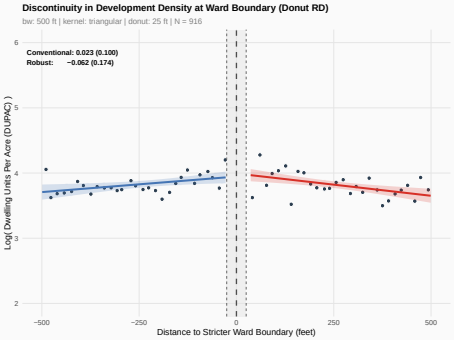
Donut RD: Excluding Parcels Within 25ft of Border

Log(FAR)



Log(DUPAC)



Discontinuity persists after excluding parcels closest to border.

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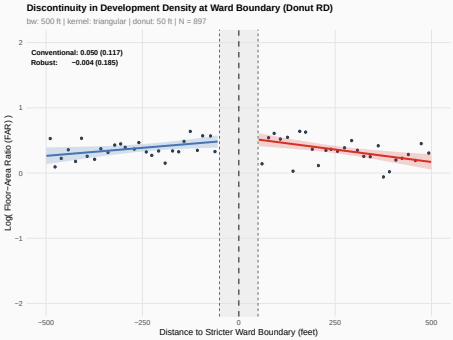
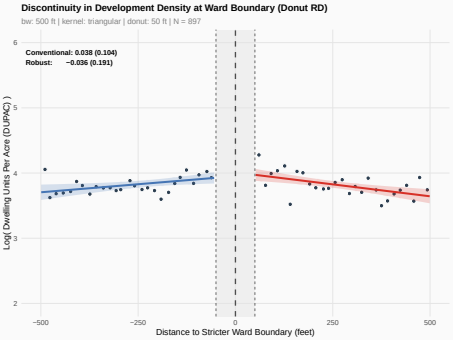
Donut RD: Excluding Parcels Within 50ft of Border

Log(FAR)



Log(DUPAC)



Discontinuity persists after excluding parcels closest to border.

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Bandwidth Sensitivity: 250ft

	ln(FAR)	ln(DUPAC)	ln(Units)
	(1)	(2)	(3)
Strictness Score	-0.24	-0.26	-0.33
	(0.14)	(0.16)	(0.22)
Observations	388	388	388
Dep. Var. Mean	2.01	62.48	11.82
Ward Pairs	89	89	89
Ward-Pair × Year FE	✓	✓	✓

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Bandwidth Sensitivity: 1000ft

	ln(FAR)	ln(DUPAC)	ln(Units)
	(1)	(2)	(3)
Strictness Score	0.03	0.03	0.02
	(0.04)	(0.04)	(0.04)
Observations	1,174	1,174	1,174
Dep. Var. Mean	1.78	54.18	9.51
Ward Pairs	120	120	120
Ward-Pair × Year FE	✓	✓	✓

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Donut Regression: Excluding 25ft from Border

	ln(FAR)	ln(DUPAC)	ln(Units)
	(1)	(2)	(3)
Strictness Score	-0.15**	-0.10	-0.11*
	(0.06)	(0.06)	(0.06)
Observations	654	654	654
Dep. Var. Mean	1.88	57.75	10.16
Ward Pairs	106	106	106
Ward-Pair × Year FE	✓	✓	✓

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Donut Regression: Excluding 50ft from Border

	ln(FAR)	ln(DUPAC)	ln(Units)
	(1)	(2)	(3)
Strictness Score	-0.15**	-0.10	-0.11*
	(0.06)	(0.06)	(0.06)
Observations	640	640	640
Dep. Var. Mean	1.88	57.86	10.26
Ward Pairs	106	106	106
Ward-Pair × Year FE	✓	✓	✓

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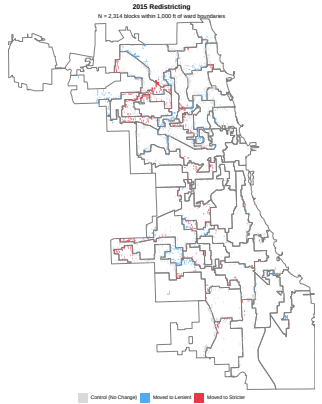
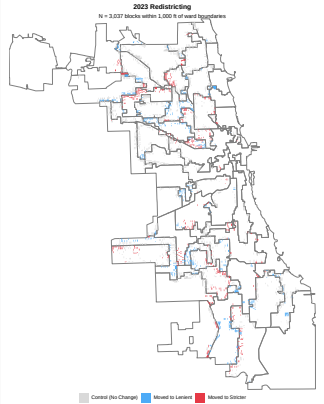
Citywide Treatment and Control Blocks

2015 Redistricting



2023 Redistricting



Red = switched to stricter; Blue = switched to lenient; Gray = control (no switch)

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Treatment and Control Block Details

	To Stricter	To Lenient	Total Treated	Control
2015 Cohort				
N Blocks	5,696	5,973	11,669	24,342
Controls per Treated		—		2.09
Mean ΔStrictness	$+0.66$	-0.58	$+0.02$	0.00
2023 Cohort				
N Blocks	3,721	4,480	8,201	23,446
Controls per Treated		—		2.86
Mean ΔStrictness	$+0.87$	-0.72	$+0.00$	0.00

Treatment defined as blocks that switched aldermen during redistricting.
Control blocks are near ward boundaries but did not switch.

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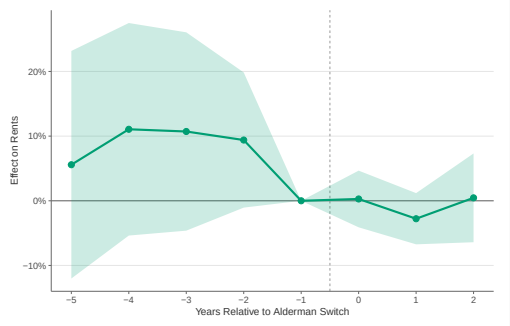
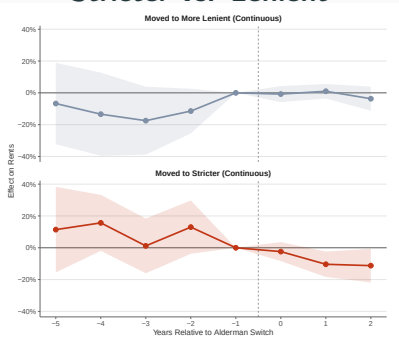
Rent Event Study: 500ft Bandwidth

Aggregate Treatment Effect



Stricter vs. Lenient



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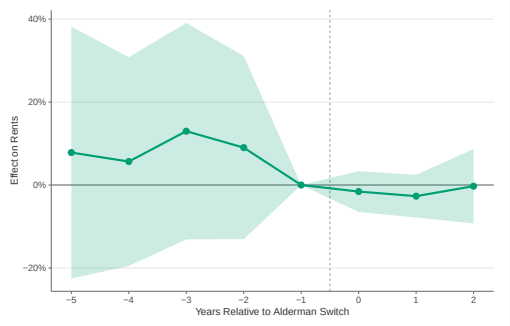
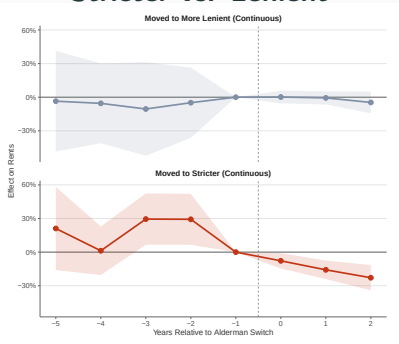
Rent Event Study: Without Hedonic Controls

Aggregate Treatment Effect



Stricter vs. Lenient



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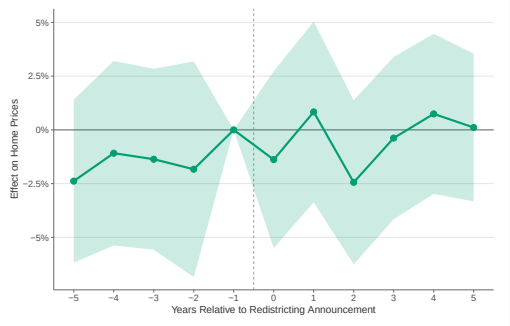
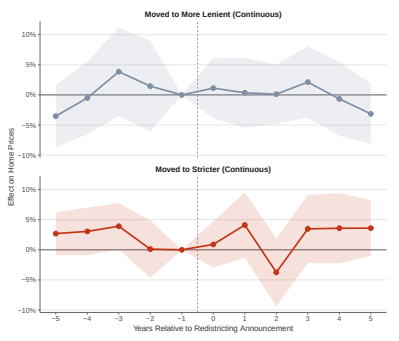
Home Price Event Study: 500 ft Bandwidth

Aggregate Treatment Effect



Stricter vs. Lenient



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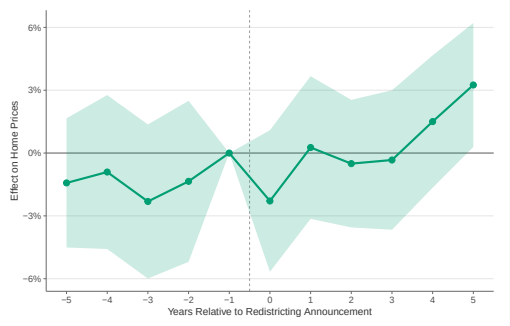
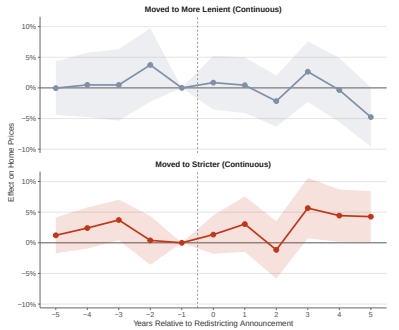
Home Price Event Study: Without Hedonic Controls

Aggregate Treatment Effect



Stricter vs. Lenient



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Covariate Balance at Ward Borders

	Mean by Group			Stricter vs Lenient	
Variable	Stricter	Lenient	Control	Diff	p-value
Sale Price ($)	$375,060	$370,964	$371,456	$4,097	0.726
N Sales	1.32	1.29	1.32	0.04	0.166
Dist to Boundary (ft)	451	457	507	-5	0.616
N Blocks	1251	1303	4133	–	–

Notes: Balance measured at $t = -1$ (one year before redistricting).

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